



Title:

**Windows Event Log Analysis Using Command Prompt
(CMD)**

Subtitle:

**A Practical Investigation of Windows Event Logs Using Native
Windows Command-Line Utilities**

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Table of Contents

- 1. Introduction**
- 2. Executive Summary**
- 3. Objectives**
- 4. Scope of Analysis**
- 5. Tools Used**
- 6. Methodology**
- 7. Findings**
- 8. Recommendations**
- 9. Lessons Learned**
- 10. Conclusion**

Introduction

Windows Event Logs are a primary source of forensic and operational evidence during cybersecurity investigations. They record authentication attempts, system events, application activities, account management operations, and administrative actions performed on a Windows system.

This project demonstrates a structured approach to Windows Event Log analysis using the native **Windows Event Utility (wevtutil)** through the Command Prompt (CMD). The investigation focuses on identifying key event logs, querying specific Event IDs, and understanding how these events contribute to security monitoring and incident investigation.

The project emphasizes practical command-line techniques commonly used by Security Operations Center (SOC) analysts to retrieve, review, and interpret Windows security events.

Executive Summary

This report documents a practical Windows Event Log analysis exercise performed using native Windows command-line tools. The investigation was conducted by querying the Windows Security, System, and Application Event Logs with the wevtutil utility.

The practical exercises focused on identifying Windows Event Logs, examining successful and failed authentication events, investigating privileged account activity, and reviewing user account management events.

The report documents the methodology used throughout the investigation, explains the purpose of each command executed, and provides space for screenshots and observed outputs. Because this report is designed as a practical laboratory exercise, no conclusions regarding malicious activity are presented unless supported by the analyst's observed results.

Objectives

The objectives of this investigation were to:

- Understand the structure of Windows Event Logs.

- Identify the primary Windows Event Logs used during security investigations.
- Learn how to query Windows Event Logs using the wevtutil command.
- Investigate authentication-related events.
- Review account management activities.
- Understand the significance of common Windows Security Event IDs.
- Document a repeatable workflow for Windows log analysis.
- Develop practical experience using native Windows forensic tools.

Scope of Analysis

The scope of this investigation was limited to Windows Event Logs accessible through the Windows Event Utility (wevtutil).

The analysis included:

- Security Event Log
- System Event Log
- Application Event Log

The investigation focused on:

- Event Log identification
- Successful logon events
- Failed logon events
- User account activity
- Administrator account authentication
- User account creation events

No external Security Information and Event Management (SIEM) platforms or third-party forensic tools were used as part of this exercise.

Windows Event Log Analysis

Tools Used

The following native Windows utility was used throughout the investigation:

- Windows Command Prompt (CMD)
- Windows Event Utility (wevtutil)

No additional forensic or third-party log analysis tools were documented for this investigation.

Methodology

The investigation followed a structured workflow commonly used during Windows log analysis.

Identify Available Event Logs

Identify the Security Event Log

Identify the System Event Log

Identify the Application Event Log

Query Successful Logon Events (Event ID 4624)

Query Failed Logon Events (Event ID 4625)

Review Successful Logon Accounts

Review Failed Logon Accounts

Review Administrator Account Activity

Event Analysis

The practical investigation focused on the following Windows Security Event IDs:

Event ID Description

4624	Successful Logon
4625	Failed Logon
4720	User Account Created

The Security, System, and Application logs were queried using wevtutil to retrieve relevant event records. Additional commands were documented for future investigation of account deletion, group membership changes, password management events, PowerShell activity, and other administrative events.

Findings

Question 1

What event logs are available on the Windows system?

Objective

Before analyzing any logs, you need to know which logs exist on the Windows system. Windows stores logs for different purposes (Security, System, Application, PowerShell, etc.).

Command: wevtutil el

What does this command mean?

Let me break it down:

- wevtutil = **Windows Event Utility**. This is a built-in Windows command-line tool used to manage Event Logs.
- el = **Enumerate Logs** (list all available event logs).

So the command simply means:

"Show me every Event Log available on this Windows computer."

Why is this important?

As a SOC analyst, you first need to identify your available evidence. You cannot investigate a log that doesn't exist.

How to answer Question 1

You don't need to list every log.

Instead, mention the major ones:

- Application
- Security
- System
- Setup

- Windows PowerShell
- Microsoft-Windows-PowerShell/Operational

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.22000.2538]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>wevtutil el
AMSI/Debug
Analytic
Application
CxAudioSvcLog
DebugChannel
DirectShowFilterGraph
DirectShowPluginControl
Els_Hyphenation/Analytic
EndpointMapper
FirstUXPerf-Analytic
ForwardedEvents
HardwareEvents
IHM_DebugChannel
Intel-GFX-Info/Application
Intel-GFX-Info/System
Intel-SST-BUS/Debug
Intel-SST-OED/IntcOED_ETW_Debug
Intel-iaLPSS-GPIO/Analytic
Intel-iaLPSS-I2C/Analytic
Intel-iaLPSS2-GPIO2/Debug
Intel-iaLPSS2-GPIO2/Performance
Intel-iaLPSS2-I2C/Debug
Intel-iaLPSS2-I2C/Performance
IntelAudioServiceLog
Internet Explorer
Key Management Service
MF_MediaFoundationDeviceMFT
MF_MediaFoundationDeviceProxy
MF_MediaFoundationFrameServer
MediaFoundationVideoProc
MediaFoundationVideoProcD3D
MediaFoundationAsyncWrapper
MediaFoundationContentProtection
MediaFoundationDS
MediaFoundationDeviceProxy
MediaFoundationMP4
```

```
Administrator: Command Prompt
Microsoft-Windows-CertificateServicesClient-Lifecycle-System/Operational
Microsoft-Windows-CertificateServicesClient-Lifecycle-User/Operational
Microsoft-Windows-Cleanmgr/Diagnostic
Microsoft-Windows-CleanTypeTextTuner/Diagnostic
Microsoft-Windows-CloudStore/Debug
Microsoft-Windows-CloudStore/Operational
Microsoft-Windows-CmiSetup/Analytic
Microsoft-Windows-CodeIntegrity/Operational
Microsoft-Windows-CodeIntegrity/Verbose
Microsoft-Windows-ComDlg32/Analytic
Microsoft-Windows-ComDlg32/Debug
Microsoft-Windows-Compat-Appraiser/Analytic
Microsoft-Windows-Compat-Appraiser/Operational
Microsoft-Windows-Containers-BindFlt/Debug
Microsoft-Windows-Containers-BindFlt/Operational
Microsoft-Windows-Containers-Wcifs/Debug
Microsoft-Windows-Containers-Wcifs/Operational
Microsoft-Windows-CoreApplication/Diagnostic
Microsoft-Windows-CoreApplication/Operational
Microsoft-Windows-CoreApplication/Tracing
Microsoft-Windows-CoreSystem-SmsRouter-Events/Debug
Microsoft-Windows-CoreSystem-SmsRouter-Events/Operational
Microsoft-Windows-CoreWindow/Analytic
Microsoft-Windows-CoreWindow/Debug
Microsoft-Windows-CorruptedFileRecovery-Client/Operational
Microsoft-Windows-CorruptedFileRecovery-Server/Operational
Microsoft-Windows-Crashdump/Operational
Microsoft-Windows-CredUI/Diagnostic
Microsoft-Windows-Crypto-BCRYPT/Analytic
Microsoft-Windows-Crypto-CNG/Analytic
Microsoft-Windows-Crypto-DPAPI/BackUpKeySvc
Microsoft-Windows-Crypto-DPAPI/Debug
Microsoft-Windows-Crypto-DPAPI/Operational
Microsoft-Windows-Crypto-DSEnh/Analytic
Microsoft-Windows-Crypto-NCrypt/Operational
Microsoft-Windows-Crypto-RNG/Analytic
Microsoft-Windows-Crypto-RSAEnh/Analytic
Microsoft-Windows-D3DI9Level9/Analytic
Microsoft-Windows-D3DI9Level9/PerfTiming
Microsoft-Windows-DAL-Provider/Analytic
```

The above screenshot show the available logs using the command wevtutil el

Identify the Security Log

Question 2

Which log contains authentication events?

Objective

The objective of this exercise is to identify which Windows Event Log stores authentication-related activities, such as:

- Successful logons
- Failed logons
- Account lockouts
- Password changes
- User creation
- Privilege escalation

As a SOC analyst, the **Security** log is usually the first place you investigate after receiving an alert about suspicious login activity.

Command: wevtutil gl Security

Understanding the Command

Let me break it down:

wevtutil

The Windows Event Utility. It allows you to manage and inspect Windows Event Logs directly from Command Prompt.

gl stands for:

Get Log

Instead of listing all logs, it retrieves detailed information about a specific log.

Security

This is the name of the log we want to inspect.

Putting it together:

wevtutil gl Security

Means:

"Show detailed information about the Windows Security Event Log."

Why Do We Use This Command?

Before investigating authentication events, you need to confirm that:

- The Security log exists.
- It is enabled.
- Windows is currently writing events to it.
- It has enough storage allocated.
- It isn't full or disabled.

This is an important verification step in a professional investigation.

Expected Output

You will see information similar to:

Name: Security

Enabled: true

type: Admin

owningPublisher:

Channelaccess:

Logging:

maximumSize: 20971520

Your result may likely be different.

```
Administrator: Command Prompt
C:\Windows\system32>wevtutil gl Security
name: Security
enabled: true
type: Admin
owningPublisher:
isolation: Custom
channelAccess: 0:BAG:SYD:(A;;0xf0005;;;SY)(A;;0x5;;;BA)(A;;0x1;;;S-1-5-32-573)
logging:
  logFileName: %SystemRoot%\System32\Winevt\Logs\Security.evtx
  retention: false
  autoBackup: false
  maxSize: 20971520
publishing:
  fileMax: 1
C:\Windows\system32>
```

Question 3

Which log records system startup and shutdown events?

Objective

Before I begin investigating system events, I need to identify the Windows log responsible for recording operating system activities.

Unlike the **Security** log, which focuses on authentication and security-related events, Windows uses the **System** log to record events generated by the operating system itself? These include system startups, shutdowns, driver initialization, hardware changes, service events, and other operating system activities.

As a cybersecurity analyst, understanding the purpose of each Windows Event Log allows me to quickly identify the correct source of evidence during an investigation.

Command: wevtutil gl System

Understanding the Command

Let me break down the command.

wevtutil

This is the **Windows Event Utility**, a built-in command-line tool used to manage and inspect Windows Event Logs.

The `gl` option stands for **Get Log**.

Instead of displaying the events inside the log, it retrieves detailed information about a specific Event Log.

System

This specifies the log I want to inspect.

Putting everything together:

```
wevtutil gl System
```

means:

Display detailed information about the Windows System Event Log.

Why Am I Running This Command?

As an analyst, I don't immediately start searching for events. I first verify that the log exists and is operational.

This command helps me answer questions such as:

- Does the System log exist?
- Is it enabled?
- Is Windows currently writing events into it?
- How many records does it contain?
- What is its maximum storage capacity?

These checks help ensure that the evidence source is available before beginning the investigation.

Expected Output

The output will look similar to this:

```
name: System
```

enabled: true

type: Admin

maximumSize: 20971520

Your values will differ depending on your system.

```
C:\Windows\system32>wevtutil gl System
name: System
enabled: true
type: Admin
owningPublisher:
isolation: System
channelAccess: O:BAG:SYD:(A;;0xf0007;;;SY)(A;;0x7;;;BA)(A;;0x3;;;BO)(A;;0x5;;;SO)(A;;0x1;;;IU)(A;;0x3;;;SU)(A;;0x1;;;S-1-5-3)(A;;0x2;;;S-1-5-33)(A;;0x1;;;S-1-5-32-573)
logging:
  logFileName: %SystemRoot%\System32\Winevt\Logs\System.evtx
  retention: false
  autoBackup: false
  maxSize: 20971520
publishing:
  fileMax: 1
C:\Windows\system32>
```

Understanding the Output

Name

System

This confirms that I am examining the System Event Log.

Enabled

true

This indicates that Windows is actively recording system events.

Maximum Size

Shows the maximum storage space allocated to the System log before Windows begins applying its retention policy.

Why is the System Log Important?

The System log records events generated by Windows itself.

Some examples include:

- System startup
- System shutdown
- Unexpected shutdowns
- Driver loading
- Hardware failures
- Disk errors
- Service startup failures
- Network adapter events

During incident response, these events help determine whether suspicious activity is related to operating system failures, hardware problems, or malicious actions.

How I Answered the Question

Question

Which log records system startup and shutdown events?

Answer

The **Windows System Event Log** records operating system events, including system startups, shutdowns, hardware events, driver initialization, and service-related activities. This log is commonly analyzed during incident investigations to determine the operational state of a Windows system.

Question 4

Which log records application-related events?

Objective

In this exercise, I want to identify the Windows Event Log responsible for recording events generated by applications installed on the system.

Unlike the **Security** log, which records authentication and audit events, and the **System** log, which records operating system activities, the **Application** log stores events generated by software applications such as web servers, databases, antivirus programs, productivity software, and custom applications?

As a cybersecurity analyst, reviewing the Application log helps identify application failures, crashes, errors, warnings, and unusual behavior that may indicate software issues or potential security concerns.

Command: wevtutil gl Application

Understanding the Command

Let's break the command down.

wevtutil

This is the **Windows Event Utility**, used to query and manage Windows Event Logs directly from the Command Prompt.

gl

The gl option means **Get Log**.

It retrieves detailed information about a specific Event Log rather than displaying the events stored inside it.

Application

This tells Windows that I want to inspect the **Application Event Log**.

Together, the command means:

Display detailed information about the Windows Application Event Log.

Why Am I Running This Command?

Before investigating application events, I first verify that the Application log is available and functioning correctly.

This command helps me confirm:

- The log exists.
- The log is enabled.
- Windows is actively recording application events.
- The total number of recorded events.
- The storage capacity allocated to the log.

This verification step ensures that the evidence source is available before performing further analysis.

Expected Output

The output will look similar to this:

name: Application

enabled: true

type: Admin

maximumSize: 20971520

The actual values on your system may differ.

Understanding the Output

Name

Application

This confirms that I am viewing the Application Event Log.

Enabled

true

This indicates that Windows is currently recording application events.

Maximum Size

Indicates the maximum amount of storage allocated to the Application log before Windows begins applying its retention policy.

```
Administrator: Command Prompt
C:\Windows\system32>wevtutil gl application
name: application
enabled: true
type: Admin
owningPublisher:
isolation: Application
channelAccess: 0:BAG:SYD:(A;;0x2;;;S-1-15-2-1)(A;;0x2;;;S-1-15-3-1024-3153509613-960666767-3724611135-2725662640-12138253-543910227-1950414635-4190290187)(A;;0xf0007;;;SY)(
A;;0x7;;;BA)(A;;0x7;;;SO)(A;;0x3;;;IU)(A;;0x3;;;SU)(A;;0x3;;;S-1-5-3)(A;;0x3;;;S-1-5-33)(A;;0x1;;;S-1-5-32-573)
logging:
  logFileName: %SystemRoot%\System32\Winevt\Logs\Application.evtx
  retention: false
  autoBackup: false
  maxSize: 20971520
publishing:
  fileMax: 1
C:\Windows\system32>
```

How I Answered the Question

Question

Which log records application-related events?

Answer

The **Windows Application Event Log** records events generated by installed applications and services. It stores information such as application errors, warnings, crashes, and informational messages. During cybersecurity investigations, this log helps analysts identify software-related issues that may contribute to or explain suspicious system behavior.

Question 5

How many successful logon events (Event ID 4624) were recorded?

Objective

The objective of this exercise is to determine the number of successful logon events recorded in the Windows Security Event Log.

A successful logon indicates that Windows authenticated a user, service, or process and granted access to the system or a network resource. Monitoring these events helps establish a baseline of normal activity and identify unusual authentication patterns.

As a cybersecurity analyst, reviewing successful logon events allows me to understand who accessed the system, when the access occurred, and how authentication was performed.

Understanding Event ID 4624

Windows assigns a unique **Event ID** to every type of event it records.

Event ID 4624 represents:

An account was successfully logged on.

This event is generated whenever Windows successfully authenticates a user or service.

Examples include:

- A user signs in locally.
- A user connects through Remote Desktop (RDP).
- A service account starts a service.
- A scheduled task runs.
- A network share is accessed successfully.

It is important to note that **a successful logon does not automatically indicate suspicious activity**. These events occur frequently during normal system operation. Analysts must interpret them in context.

Command

```
wevtutil qe Security /q:"*[System[(EventID=4624)]]" /f:text
```

Understanding the Command

Let's examine each part of the command.

wevtutil

The Windows Event Utility used to query and manage Windows Event Logs.

qe

The qe option stands for **Query Events**.

Unlike `gl`, which retrieves information about a log, `qe` retrieves the actual events stored within a log.

Security

Specifies that I want to search the **Security Event Log**.

/q

This stands for **Query**.

It allows me to filter events instead of displaying every event in the log.

"*[System[(EventID=4624)]]"

This is an XPath query.

It instructs Windows to return **only events whose Event ID equals 4624**.

Without this filter, thousands of unrelated events would be displayed.

/f:text

This tells Windows to display the results in **plain text**, making them easier to read from the Command Prompt.

What Does the Command Do?

In simple terms, this command tells Windows:

"Search the Security Event Log and display every event whose Event ID is 4624 (successful logon), then show the results as readable text."

Expected Output

You may see output similar to:

Your output will vary depending on your system.

Event[2166]

Log Name: Security

Source: Microsoft-Windows-Security-Auditing

Date: 2026-07-09T19:54:48.1770000Z

Event ID: 4624

Task: Logon

Level: Information

Opcode: Info

Keyword: Audit Success

User: N/A

User Name: N/A

Computer: DESKTOP-CNV4E20

Description:

An account was successfully logged on.

```
Event[2166]
Log Name: Security
Source: Microsoft-Windows-Security-Auditing
Date: 2026-07-09T19:54:48.1770000Z
Event ID: 4624
Task: Logon
Level: Information
Opcode: Info
Keyword: Audit Success
User: N/A
User Name: N/A
Computer: DESKTOP-CNV4E20
Description:
An account was successfully logged on.

Subject:
  Security ID: S-1-5-18
  Account Name: DESKTOP-CNV4E20$
  Account Domain: WORKGROUP
  Logon ID: 0x3E7

Logon Information:
  Logon Type: 5
  Restricted Admin Mode: -
  Virtual Account: No
  Elevated Token: Yes

Impersonation Level: Impersonation

New Logon:
  Security ID: S-1-5-18
  Account Name: SYSTEM
  Account Domain: NT AUTHORITY
  Logon ID: 0x3E7
  Linked Logon ID: 0x0
  Network Account Name: -
  Network Account Domain: -
  Logon GUID: {00000000-0000-0000-0000-000000000000}
```

Account Name

Identifies the user or service account that successfully authenticated.

Logon Type

Indicates how the authentication occurred.

Common values include:

Logon Type Meaning

2	Interactive (local keyboard login)
3	Network login
4	Batch job
5	Service account
7	Unlock workstation
10	Remote Desktop (RDP)
11	Cached credentials

Understanding the logon type helps determine whether the login is expected or requires further investigation.

Counting the Events

The previous command displays all matching events. To count them, use:

```
wevtutil qe Security /q:"*[System[(EventID=4624)]]" /f:text | find /c "EventID: 4624"
```

```
C:\Windows\system32>wevtutil qe Security /q:"*[System[(EventID=4624)]]" /f:text | find /c "Event ID: 4624"
2287
C:\Windows\system32>
```

Understanding the Second Command

The vertical bar (|) is called a **pipe**.

It sends the output of the first command directly into the second command.

The second command:

```
find /c "Event ID: 4624"
```

Counts how many times the text "**Event ID: 4624**" appears.

The result is the total number of successful logon events which is 2287

Question 6

How many failed logon events (Event ID 4625) were recorded?

Objective

The objective of this exercise is to determine the number of **failed authentication attempts** recorded in the Windows Security Event Log.

A failed logon occurs when Windows denies access because the authentication process was unsuccessful. This may happen when an incorrect username or password is supplied, an account is locked, the account is disabled, or a logon is attempted using an unauthorized method.

As a Cybersecurity Analyst, monitoring failed logon events is essential because repeated authentication failures may indicate password guessing, brute-force attacks, password spraying, or unauthorized attempts to access the system.

Understanding Event ID 4625

Every Windows security event has a unique **Event ID**.

Event ID 4625 means:

An account failed to log on.

Windows generates this event whenever an authentication attempt is unsuccessful.

Common reasons include:

- Incorrect password
- Incorrect username
- Disabled account
- Locked account
- Expired password
- Expired account
- Invalid logon method

A single failed logon is not necessarily suspicious. However, **multiple failed logons in a short period**, especially from the same account or IP address, may require further investigation.

Command

```
wevtutil qe Security /q:"*[System[(EventID=4625)]]" /f:text
```

Understanding the Command

Let's break it down.

wevtutil

The Windows Event Utility used to manage and query Windows Event Logs.

qe

Query Events

Retrieves events stored inside a specific Event Log.

Security

Specifies that I want to search the **Security Event Log**, where authentication events are stored.

/q

Applies a query filter.

Instead of displaying every event, Windows returns only the events that match the filter.

"*[System[(EventID=4625)]]"

This XPath filter tells Windows:

"Display only events who's Event ID equals **4625**."

/f:text

Displays the results in plain text, making them easy to read from the Command Prompt.

What Does This Command Do?

In simple terms, this command tells Windows:

"Search the Security Event Log and display every failed logon event."

```
C:\Windows\system32>wevtutil qe Security /q:"*[System[(EventID=4625)]]" /f:text
Event[0]
  Log Name: Security
  Source: Microsoft-Windows-Security-Auditing
  Date: 2026-06-28T18:28:34.2140000Z
  Event ID: 4625
  Task: Logon
  Level: Information
  Opcode: Info
  Keyword: Audit Failure
  User: N/A
  User Name: N/A
  Computer: DESKTOP-CNV4E20
  Description:
An account failed to log on.

Subject:
  Security ID: S-1-5-18
  Account Name: DESKTOP-CNV4E20$
  Account Domain: WORKGROUP
  Logon ID: 0x3E7

Logon Type: 2

Account For Which Logon Failed:
  Security ID: S-1-0-0
  Account Name: HP
  Account Domain: DESKTOP-CNV4E20

Failure Information:
  Failure Reason: Unknown user name or bad password.
  Status: 0xC000006D
  Sub Status: 0xC000006A

Process Information:
  Caller Process ID: 0x700
  Caller Process Name: C:\Windows\System32\svchost.exe

Network Information:
```

Counting the Failed Logons

To count the total number of failed logon events, use:

```
wevtutil qe Security /q:"*[System[(EventID=4625)]]" /f:text | find /c "Event ID: 4625"
```

```
C:\Windows\system32>wevtutil qe Security /q:"*[System[(EventID=4625)]]" /f:text | find /c "Event ID: 4625"
6
C:\Windows\system32>
```

Understanding the Pipe (|)

The **pipe** sends the output of the first command directly into the second command.

```
find /c "Event ID: 4625"
```

counts how many times the text "**Event ID: 4625**" appears.

The result is the total number of failed logon events which occurred 6 times.

Question 7

Which account generated the most successful logons?

Objective

The objective of this exercise is to determine which Windows user account recorded the highest number of successful logon events.

A successful logon does not necessarily indicate suspicious activity. However, identifying the most active account helps establish a baseline of normal user behavior. During incident response, this information can also reveal service accounts, administrative accounts, or user accounts that require further investigation due to unusually high authentication activity.

As a Cybersecurity Analyst, understanding user authentication patterns is an important part of detecting anomalies and investigating potential security incidents.

Understanding Event ID 4624

As discussed in the previous exercise, **Event ID 4624** represents:

An account was successfully logged on.

Each Event ID 4624 contains important information such as:

- Account Name
- Logon Type
- Authentication Package
- Source Network Address
- Logon Process
- Time Created

For this exercise, I am interested in the **Account Name** field.

Command

```
wevtutil ql Security /q:"*[System[(EventID=4624)]]" /f:text | findstr "Account Name"
```

Understanding the Command

Let's break it down.

wevtutil

Queries Windows Event Logs.

ql

Query Events.

Security

Searches the Windows Security Event Log.

/q

Applies a filter.

```
"*[System[(EventID=4624)]]"
```

Returns only successful logon events.

/f:text

Displays the events in readable text.

|

The pipe (|) sends the output of the first command into another command.

findstr

findstr searches text for a specific word or phrase.

In this case,

findstr "Account Name"

means:

"Show me only the lines that contain **Account Name**."

Instead of displaying hundreds of lines for each event, Windows displays only the usernames associated with successful logons.

```

User Name: N/A
  Account Name:      DESKTOP-CNV4E20$
  Account Domain:    WORKGROUP
  Virtual Account:    No
  Account Name:      SYSTEM
  Account Domain:    NT AUTHORITY
  Network Account Name: -
  Network Account Domain: -
  Process Name:      C:\Windows\System32\services.exe
  Workstation Name:  -
  Package Name (NTLM only): -
Log Name: Security
User Name: N/A
  Account Name:      DESKTOP-CNV4E20$
  Account Domain:    WORKGROUP
  Virtual Account:    No
  Account Name:      SYSTEM
  Account Domain:    NT AUTHORITY
  Network Account Name: -
  Network Account Domain: -
  Process Name:      C:\Windows\System32\services.exe
  Workstation Name:  -
  Package Name (NTLM only): -
Log Name: Security
User Name: N/A
  Account Name:      DESKTOP-CNV4E20$
  Account Domain:    WORKGROUP
  Virtual Account:    No
  Account Name:      SYSTEM
  Account Domain:    NT AUTHORITY
  Network Account Name: -
  Network Account Domain: -
  Process Name:      C:\Windows\System32\services.exe
  Workstation Name:  -
  Package Name (NTLM only): -
Log Name: Security
User Name: N/A
  Account Name:      DESKTOP-CNV4E20$
  Account Domain:    WORKGROUP
  Virtual Account:    No

```

Your output will depend on the activity recorded on your system.

Why Isn't This Enough?

This command lists the account names, but it **does not tell me which account appears most frequently**.

To answer the assignment question, I need to **count** how many times each account appears.

Improved Command

```
wevtutil qe Security /q:"*[System[(EventID=4624)]]" /f:text | findstr "Account  
Name" | sort
```

This sorts the account names alphabetically, making them easier to review.

[illegible]

Even Better (PowerShell)

Command Prompt is limited when it comes to grouping and counting text. If you later use PowerShell, you can determine the most frequent account much more easily.

However, because this project focuses on **CMD**, I will manually review the sorted list and identify the account that appears most often.

Question 8

Which account generated failed logon attempts?

Objective

The objective of this exercise is to identify the user account(s) associated with failed authentication attempts recorded in the Windows Security Event Log.

When Windows rejects a login attempt, it records **Event ID 4625**. By examining these events, I can determine which accounts experienced failed logons and whether the failures represent normal user mistakes or potentially malicious activity.

As a Cybersecurity Analyst, identifying accounts associated with failed logons is an important step in detecting password guessing, brute-force attacks, disabled accounts, expired passwords, or other authentication-related issues.

Understanding Event ID 4625

Windows assigns **Event ID 4625** whenever a logon attempt fails.

This event contains valuable information including:

- Account Name
- Failure Reason
- Logon Type
- Source Network Address
- Workstation Name
- Time Created

For this exercise, my primary focus is the **Account Name** field.

Command

```
wevtutil qe Security /q:"*[System[(EventID=4625)]]" /f:text | findstr "Account Name"
```

Understanding the Command

Let's examine each part of the command.

wevtutil

The Windows Event Utility used to query Windows Event Logs.

qe

Query Events

Retrieves events from a specified Event Log.

Security

Specifies that the search should be performed in the Windows Security Event Log.

/q

Applies a query filter.

"*[System[(EventID=4625)]]"

Returns only failed logon events.

/f:text

Displays the events in plain text for easier reading.

|

The pipe (|) passes the output of the first command into another command.

findstr

Searches for lines containing a specific word or phrase.

In this case,

`findstr "Account Name"`

extracts only the usernames from the failed logon events.

```
Log Name: Security
User Name: N/A
Account Name: DESKTOP-CNV4E20$
Account Domain: WORKGROUP
Account For Which Logon Failed:
Account Name: HP
Account Domain: DESKTOP-CNV4E20
Caller Process Name: C:\Windows\System32\svchost.exe
Workstation Name: DESKTOP-CNV4E20
Package Name (NTLM only): -
Log Name: Security
User Name: N/A
Account Name: DESKTOP-CNV4E20$
Account Domain: WORKGROUP
Account For Which Logon Failed:
Account Name: HP
Account Domain: DESKTOP-CNV4E20
Caller Process Name: C:\Windows\System32\svchost.exe
Workstation Name: DESKTOP-CNV4E20
Package Name (NTLM only): -
Log Name: Security
User Name: N/A
Account Name: DESKTOP-CNV4E20$
Account Domain: WORKGROUP
Account For Which Logon Failed:
Account Name: HP
Account Domain: DESKTOP-CNV4E20
Caller Process Name: C:\Windows\System32\svchost.exe
Workstation Name: DESKTOP-CNV4E20
Package Name (NTLM only): -
Log Name: Security
User Name: N/A
Account Name: DESKTOP-CNV4E20$
Account Domain: WORKGROUP
Account For Which Logon Failed:
Account Name: HP
Account Domain: DESKTOP-CNV4E20
Caller Process Name: C:\Windows\System32\svchost.exe
Workstation Name: DESKTOP-CNV4E20
```

What Does This Command Do?

In simple terms, this command tells Windows:

"Display every failed logon event and show only the account names involved."

Instead of reviewing every detail of each event, I can quickly focus on the affected accounts.

The names shown will depend on the activity recorded on your system.

Why Is This Important?

As a security analyst, I want to know:

- Which accounts are experiencing failed logon attempts?
- Are the failures concentrated on one account?

- Is a privileged account, such as **Administrator**, being targeted?
- Are there multiple failed attempts against the same account, which could indicate password guessing or brute-force activity? A single failed logon may simply be a typing mistake. However, repeated failures involving the same account deserve closer examination.

Question 9

Were there any logon attempts using the Administrator account?

Objective

The objective of this exercise is to determine whether the built-in **Administrator** account was used during authentication attempts recorded in the Windows Security Event Log.

Administrative accounts have elevated privileges and are often targeted by attackers during password guessing, brute-force attacks, and privilege escalation attempts. By reviewing authentication events involving the Administrator account, I can determine whether the account was used successfully, unsuccessfully, or not at all during the available logging period.

As a Cybersecurity Analyst, monitoring privileged account activity is an essential part of detecting unauthorized access and protecting critical systems.

Why Is the Administrator Account Important?

The built-in **Administrator** account has unrestricted access to the Windows operating system.

If an attacker gains access to this account, they may be able to:

- Install malicious software
- Create or delete user accounts
- Disable antivirus software
- Modify security settings
- Access sensitive files
- Control the entire system

For this reason, security teams closely monitor all authentication attempts involving privileged accounts.

Command

```
wevtutil qe Security /q:"*[System[(EventID=4624)]]" /f:text | findstr /i "Administrator"
```

Understanding the Command

wevtutil

Queries Windows Event Logs.

qe

Retrieves events from the log.

Security

Searches the Windows Security Event Log.

/q

Applies a filter.

"*[System[(EventID=4624)]]"

Returns only **successful logon events**.

/f:text

Displays the events in plain text.

findstr

Searches the output for a specific word.

/i

Makes the search **case-insensitive**.

This means Windows will find:

Administrator
administrator
ADMINISTRATOR
"Administrator"

Searches for every successful logon involving the Administrator account.

What Does This Command Do?

In simple terms:

"Search all successful logon events and display only those involving the Administrator account."

Question 10

Were any new user accounts created? (Event ID 4720)

Objective

The objective of this exercise is to determine whether any new user accounts were created on the Windows system during the available logging period.

Windows records the creation of a new user account as **Event ID 4720** in the **Security Event Log**. Monitoring account creation events is important because unauthorized accounts can be used by attackers to establish persistence, escalate privileges, or maintain long-term access to a compromised system.

As a Cybersecurity Analyst, reviewing account creation events helps identify legitimate administrative activities as well as potentially suspicious user management actions.

Understanding Event ID 4720

Windows assigns **Event ID 4720** whenever a new user account is created.

This event records information such as:

- Name of the newly created account
- Account that created the user
- Time the account was created
- Security Identifier (SID)
- Domain information

The event allows analysts to determine **who created the account, when it was created, and which account performed the action.**

Command

```
Microsoft Windows [Version 10.0.22000.2538]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>wevtutil qe Security /q:"*[System[(EventID=4720)]]" /f:text

C:\Windows\system32>
```

Understanding the Command

Let's examine each component of the command.

wevtutil

The Windows Event Utility used to query and manage Windows Event Logs.

qe

Query Events

Retrieves matching events from the selected Event Log.

Security

Specifies that the search should be performed within the Windows Security Event Log.

/q

Applies a query filter to limit the results.

```
"*[System[(EventID=4720)]]"
```

This XPath query filters the Security log and returns **only events with Event ID 4720**, which represent newly created user accounts.

```
/f:text
```

Displays the event details in plain text, making them easier to review in the Command Prompt.

What Does This Command Do?

In simple terms, the command tells Windows:

"Search the Security Event Log and display every event where a new user account was created."

This allows me to quickly identify any account creation activity without manually reviewing thousands of unrelated events.

Additional Commands for Further Investigation

Question 11

Were any user accounts deleted? (Event ID 4726)

Command

```
wevtutil qe Security /q:"*[System[(EventID=4726)]]" /f:text
```

Explanation

Searches the Security Event Log for Event ID **4726**, which records user account deletion events. This helps identify whether any accounts were removed during the investigation period.

Question 12

Were any user accounts added to privileged groups? (Event ID 4732)

Command

```
wevtutil qe Security /q:"*[System[(EventID=4732)]]" /f:text
```

Explanation

Displays events where a user account was added to a security-enabled local group, such as the **Administrators** group. Analysts review these events to detect privilege escalation.

Question 13

Were any passwords changed?

Command

```
wevtutil qe Security /q:"*[System[(EventID=4723 or EventID=4724)]]" /f:text
```

Explanation

Searches for password-related events.

- **4723** – User attempted to change a password.
- **4724** – An administrator reset another user's password.

These events are useful during account management investigations.

Question 14

How many times was the system restarted?

Command

```
wevtutil qe System /q:"*[System[(EventID=1074)]]" /f:text
```

Explanation

Displays planned restart or shutdown events initiated by a user or process.

Question 15

How many unexpected shutdowns were recorded?

Command

```
wevtutil qe System /q:"*[System[(EventID=6008)]]" /f:text
```

Explanation

Shows unexpected shutdown events. These may indicate power failures, crashes, or forced shutdowns.

Question 16

Were any critical system errors observed?

Command

```
wevtutil qe System /q:"*[System[(Level=1)]]" /f:text
```

Explanation

Displays **Critical** events recorded in the System Event Log. Analysts review these events to identify serious operating system issues.

Question 17

What was the most recent system event?

Command

```
wevtutil qe System /c:1 /rd:true /f:text
```

Explanation

Retrieves the most recent event recorded in the System Event Log.

Question 18

Was PowerShell used on the system?

Command

```
wevtutil qe Microsoft-Windows-PowerShell/Operational /c:20 /rd:true /f:text
```

Explanation

Displays recent PowerShell Operational log entries, allowing analysts to determine whether PowerShell activity occurred.

Question 19

How many PowerShell operational events were recorded?

Command

```
wevtutil qe Microsoft-Windows-PowerShell/Operational /f:text | find /c "Event ID"
```

Explanation

Counts the total number of PowerShell events recorded in the Operational log.

Question 20

Were any suspicious PowerShell commands observed?

Command

```
wevtutil qe Microsoft-Windows-PowerShell/Operational /f:text
```

Explanation

Displays PowerShell Operational events for manual review. Analysts examine the output for unusual scripts or commands that may indicate malicious activity.

Question 21

Which Event IDs appeared most frequently?

Command

```
wevtutil qe Security /f:text | findstr "Event ID" | sort
```

Explanation

Lists Event IDs found in the Security log. Analysts can review the output to identify the events that occur most often.

Question 22

Were there any repeated failed logon attempts?

Command

```
wevtutil qe Security /q:"*[System[(EventID=4625)]]" /f:text
```

Explanation

Displays all failed logon events. Analysts look for repeated attempts against the same account or from the same source.

Question 23

Was there evidence of possible brute-force activity?

Command

```
wevtutil qe Security /q:"*[System[(EventID=4625)]]" /f:text
```

Explanation

Review failed authentication events for repeated failures targeting the same account or occurring within a short time. This may indicate a brute-force attempt, but additional context is required before reaching a conclusion.

Question 24

Were there any suspicious administrative actions?

Command

```
wevtutil qe Security /f:text | findstr "4720 4726 4732 4723 4724"
```

Explanation

Searches for account creation, deletion, group membership changes, and password-related events that may indicate administrative activity.

Question 25

What security observations can be made from the available logs?

Explanation

Review the authentication events, account management activities, PowerShell logs, and system events together to identify notable patterns. Base all observations on the available evidence without making unsupported assumptions.

Recommendations

Based on the scope of this practical exercise, the following recommendations are provided:

- Continue monitoring Windows Security Event Logs for authentication-related activity.
- Regularly review privileged account usage.
- Investigate repeated failed logon attempts to determine whether they indicate password guessing or unauthorized access attempts.
- Monitor account management events such as account creation, deletion, and privilege changes.
- Preserve Windows Event Logs as evidence before conducting incident response activities.
- Consider integrating Windows Event Logs into a centralized SIEM platform for improved monitoring and correlation.

Lessons Learned

This practical exercise provided experience in using native Windows command-line tools to perform security log analysis. It reinforced the importance of understanding Windows Event Logs, identifying key Event IDs, and documenting investigative procedures in a structured and repeatable manner. The exercise also demonstrated how command-line utilities can be used to retrieve valuable security information without relying on third-party tools.

Conclusion

This report presented a structured approach to Windows Event Log analysis using the Windows Event Utility (`wevtutil`) through the Command Prompt.

The investigation successfully documented the process for identifying Windows Event Logs and querying authentication and account management events. Because the report was intentionally limited to documented procedures and observed evidence, no conclusions regarding suspicious or malicious activity were made without supporting command output.

The documented methodology provides a repeatable framework for future Windows Event Log investigations and supports the development of practical log analysis skills applicable to both academic environments and Security Operations Centers (SOCs).